TASK 8: DATABASE RELATIONSHIP ANALYSIS USING JOINS

PROJECT DETAILS

- **Name:** Kaviya S
- **Roll No:** E25AI118
- **Department:** B.Sc Computer Science with AI

SCENARIO

The bookstore **PALETTE: Pixel & Pages** stores customer, product (book), order, payment, and inventory information in separate tables. The management team wants to combine these related tables to understand customer purchases, order details, payment status, and business performance.

1. IMPLEMENT INNER JOIN

1.1 Display customer details along with their orders

```
SQL> SELECT
  2  c.Customer_ID,
  3  c.First_Name,
  4  c.Last_Name,
  5  o.Order_ID,
  6  o.Order_Date,
  7  o.Order_Status
  8  FROM Customer c
  9  INNER JOIN Orders o
 10  ON c.Customer_ID = o.Customer_ID;

CUSTOMER_ID          FIRST_NAME           LAST_NAME            ORDER_ID
-------------------- -------------------- -------------------- ----------
ORDER_DAT ORDER_STATUS
--------- --------------------
CUST-101             Ananya               Roy                  ORD-501
01-MAR-26 Delivered

CUST-101             Ananya               Roy                  ORD-502
05-MAR-26 Delivered

CUST-102             Rohan                Mehta                ORD-503
10-MAR-26 Pending


CUSTOMER_ID          FIRST_NAME           LAST_NAME            ORDER_ID
-------------------- -------------------- -------------------- ----------
ORDER_DAT ORDER_STATUS
--------- --------------------
CUST-103             Priya                Nair                 ORD-504
12-MAR-26 Shipped

CUST-105             Aarav                Sharma               ORD-505
15-MAR-26 Delivered

CUST-106             Ishita               Menon                ORD-506
16-MAR-26 Pending


CUSTOMER_ID          FIRST_NAME           LAST_NAME            ORDER_ID
-------------------- -------------------- -------------------- ----------
ORDER_DAT ORDER_STATUS
--------- --------------------
CUST-107             Kabir                Singh                ORD-507
17-MAR-26 Shipped

CUST-108             Diya                 Patel                ORD-508
18-MAR-26 Delivered

CUST-109             Arjun                Nair                 ORD-509
19-MAR-26 Delivered
```

```
CUSTOMER_ID          FIRST_NAME           LAST_NAME            ORDER_ID
-------------------- -------------------- -------------------- ----------
ORDER_DAT ORDER_STATUS
--------- --------------------
CUST-110             Meera                Rao                  ORD-510
20-MAR-26 Pending

CUST-111             Rohan                Verma                ORD-511
21-MAR-26 Delivered

CUST-112             Anika                Iyer                 ORD-512
22-MAR-26 Shipped


CUSTOMER_ID          FIRST_NAME           LAST_NAME            ORDER_ID
-------------------- -------------------- -------------------- ----------
ORDER_DAT ORDER_STATUS
--------- --------------------
CUST-113             Nikhil               Bose                 ORD-513
23-MAR-26 Pending

CUST-114             Pooja                Reddy                ORD-514
24-MAR-26 Delivered

CUST-115             Vikram               Kapoor               ORD-515
25-MAR-26 Delivered


CUSTOMER_ID          FIRST_NAME           LAST_NAME            ORDER_ID
-------------------- -------------------- -------------------- ----------
ORDER_DAT ORDER_STATUS
--------- --------------------
CUST-116             Sneha                Gupta                ORD-516
26-MAR-26 Cancelled

CUST-117             Aditya               Mishra               ORD-517
27-MAR-26 Delivered

CUST-118             Kavya                Das                  ORD-518
28-MAR-26 Shipped


CUSTOMER_ID          FIRST_NAME           LAST_NAME            ORDER_ID
-------------------- -------------------- -------------------- ----------
ORDER_DAT ORDER_STATUS
--------- --------------------
CUST-119             Rahul                Joseph               ORD-519
29-MAR-26 Pending

CUST-120             Neha                 Varma                ORD-520
30-MAR-26 Delivered
```

```
20 rows selected.
```

1.2 Display order details with payment information

```
SQL> SELECT
  2  o.Order_ID,
  3  o.Order_Date,
  4  o.Total_Amount,
  5  p.Payment_ID,
  6  p.Payment_Mode,
  7  p.Payment_Status
  8  FROM Orders o
  9  INNER JOIN Payment p
 10  ON o.Order_ID = p.Order_ID;

ORDER_ID   ORDER_DAT TOTAL_AMOUNT PAYMENT PAYMENT_MODE    PAYMENT_ST
---------- --------- ------------ ------- --------------- ----------
ORD-501    01-MAR-26         1098 PAY001  Credit Card     Completed
ORD-502    05-MAR-26         1250 PAY002  UPI             Completed
ORD-503    10-MAR-26          599 PAY003  Debit Card      Pending
ORD-504    12-MAR-26         1149 PAY004  Net Banking     Completed
ORD-505    15-MAR-26          699 PAY005  UPI             Completed
ORD-506    16-MAR-26          799 PAY006  Credit Card     Pending
ORD-507    17-MAR-26          499 PAY007  UPI             Completed
ORD-508    18-MAR-26          599 PAY008  Debit Card      Completed
ORD-509    19-MAR-26          450 PAY009  Net Banking     Completed
ORD-510    20-MAR-26          650 PAY010  UPI             Pending
ORD-511    21-MAR-26          999 PAY011  Credit Card     Completed

ORDER_ID   ORDER_DAT TOTAL_AMOUNT PAYMENT PAYMENT_MODE    PAYMENT_ST
---------- --------- ------------ ------- --------------- ----------
ORD-512    22-MAR-26          850 PAY012  Debit Card      Completed
ORD-513    23-MAR-26          550 PAY013  UPI             Pending
ORD-514    24-MAR-26          799 PAY014  Credit Card     Completed
ORD-515    25-MAR-26          899 PAY015  Net Banking     Completed
ORD-516    26-MAR-26          450 PAY016  UPI             Failed
ORD-517    27-MAR-26          699 PAY017  Credit Card     Completed
ORD-518    28-MAR-26          750 PAY018  Debit Card      Completed
ORD-519    29-MAR-26          399 PAY019  UPI             Pending
ORD-520    30-MAR-26          550 PAY020  Net Banking     Completed

20 rows selected.
```

1.3 Retrieve products purchased by customers (Books)

```
SQL> SELECT
  2  c.First_Name,
  3  c.Last_Name,
  4  b.Title
  5  FROM Customer c
  6  INNER JOIN Orders o
  7  ON c.Customer_ID = o.Customer_ID
  8  INNER JOIN Order_Details od
  9  ON o.Order_ID = od.Order_ID
 10  INNER JOIN Book b
 11  ON od.Product_ID = b.Book_ID;

FIRST_NAME           LAST_NAME
-------------------- --------------------
TITLE
--------------------------------------------------------------------------------
Priya                Nair
The Great Gatsby

Ananya               Roy
The Great Gatsby

Rohan                Mehta
Dune


FIRST_NAME           LAST_NAME
-------------------- --------------------
TITLE
--------------------------------------------------------------------------------
Ananya               Roy
Dune

Priya                Nair
Foundation

Ananya               Roy
Designing Data-Intensive Applications


FIRST_NAME           LAST_NAME
-------------------- --------------------
TITLE
--------------------------------------------------------------------------------
Aarav                Sharma
The Silent Patient

Ishita               Menon
Gone Girl

Kabir                Singh
Sherlock Holmes
```

```
FIRST_NAME           LAST_NAME
-------------------- --------------------
TITLE
--------------------------------------------------------------------------------
Diya                 Patel
Murder on the Orient Express

Arjun                Nair
Pride and Prejudice

Meera                Rao
Me Before You


FIRST_NAME           LAST_NAME
-------------------- --------------------
TITLE
--------------------------------------------------------------------------------
Rohan                Verma
Harry Potter and the Philosopher's Stone

Anika                Iyer
The Hobbit

Nikhil               Bose
Dracula


FIRST_NAME           LAST_NAME
-------------------- --------------------
TITLE
--------------------------------------------------------------------------------
Pooja                Reddy
The Shining

Vikram               Kapoor
Wings of Fire

Sneha                Gupta
The Diary of a Young Girl


FIRST_NAME           LAST_NAME
-------------------- --------------------
TITLE
--------------------------------------------------------------------------------
Rahul                Joseph
Treasure Island

Neha                 Varma
Around the World in 80 Days

Aditya               Mishra
Percy Jackson and the Lightning Thief
```

```
FIRST_NAME           LAST_NAME
-------------------- --------------------
TITLE
--------------------------------------------------------------------------------
Kavya                Das
Angels and Demons


22 rows selected.
```

2. IMPLEMENT LEFT JOIN

2.1 Display all customers including customers who have not placed orders

```
SQL> SELECT
  2  c.Customer_ID,
  3  c.First_Name,
  4  c.Last_Name,
  5  o.Order_ID
  6  FROM Customer c
  7  LEFT JOIN Orders o
  8  ON c.Customer_ID = o.Customer_ID;

CUSTOMER_ID          FIRST_NAME           LAST_NAME            ORDER_ID
-------------------- -------------------- -------------------- ----------
CUST-101             Ananya               Roy                  ORD-501
CUST-101             Ananya               Roy                  ORD-502
CUST-102             Rohan                Mehta                ORD-503
CUST-103             Priya                Nair                 ORD-504
CUST-105             Aarav                Sharma               ORD-505
CUST-106             Ishita               Menon                ORD-506
CUST-107             Kabir                Singh                ORD-507
CUST-108             Diya                 Patel                ORD-508
CUST-109             Arjun                Nair                 ORD-509
CUST-110             Meera                Rao                  ORD-510
CUST-111             Rohan                Verma                ORD-511

CUSTOMER_ID          FIRST_NAME           LAST_NAME            ORDER_ID
-------------------- -------------------- -------------------- ----------
CUST-112             Anika                Iyer                 ORD-512
CUST-113             Nikhil               Bose                 ORD-513
CUST-114             Pooja                Reddy                ORD-514
CUST-115             Vikram               Kapoor               ORD-515
CUST-116             Sneha                Gupta                ORD-516
CUST-117             Aditya               Mishra               ORD-517
CUST-118             Kavya                Das                  ORD-518
CUST-119             Rahul                Joseph               ORD-519
CUST-120             Neha                 Varma                ORD-520
CUST-104             Vikram               Seth

21 rows selected.
```

2.2 Find customers without any purchases

```
SQL> SELECT
  2  c.Customer_ID,
  3  c.First_Name,
  4  c.Last_Name
  5  FROM Customer c
  6  LEFT JOIN Orders o
  7  ON c.Customer_ID = o.Customer_ID
  8  WHERE o.Order_ID IS NULL;

CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
CUST-104             Vikram               Seth
```

2.3 Display all products including products with no sales (Books)

```
SQL> SELECT
  2  b.Book_ID,
  3  b.Title,
  4  od.Order_ID
  5  FROM Book b
  6  LEFT JOIN Order_Details od
  7  ON b.Book_ID = od.Product_ID;

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
ORDER_ID
----------
B1001
The Great Gatsby
ORD-501

B1002
Dune
ORD-501

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
ORDER_ID
----------

B1004
Designing Data-Intensive Applications
ORD-502

B1002
Dune

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
ORDER_ID
----------
ORD-503

B1001
The Great Gatsby
ORD-504

B1003
```

```
BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
ORDER_ID
----------
Foundation
ORD-504

B1005
The Silent Patient
ORD-505


BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
ORDER_ID
----------
B1006
Gone Girl
ORD-506

B1007
Sherlock Holmes
ORD-507

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
ORDER_ID
----------

B1008
Murder on the Orient Express
ORD-508

B1009
Pride and Prejudice

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
ORDER_ID
----------
ORD-509

B1010
Me Before You
ORD-510

B1011
```

```
BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
ORDER_ID
----------
Harry Potter and the Philosopher's Stone
ORD-511

B1012
The Hobbit
ORD-512


BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
ORDER_ID
----------
B1013
Dracula
ORD-513

B1014
The Shining
ORD-514

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
ORDER_ID
----------

B1015
Wings of Fire
ORD-515

B1018
The Diary of a Young Girl

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
ORDER_ID
----------
ORD-516

B1022
Percy Jackson and the Lightning Thief
ORD-517
```

```
BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
ORDER_ID
----------
Angels and Demons
ORD-518

B1019
Treasure Island
ORD-519


BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
ORDER_ID
----------
B1020
Around the World in 80 Days
ORD-520

B1023
The Da Vinci Code


BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
ORDER_ID
----------

B1016
Steve Jobs


B1017
Sapiens

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
ORDER_ID
----------


B1021
The Fault in Our Stars
```

```
26 rows selected.
```

3. IMPLEMENT RIGHT JOIN

3.1 Display all orders with customer information

```
SQL> SELECT
  2  c.Customer_ID,
  3  c.First_Name,
  4  c.Last_Name,
  5  o.Order_ID,
  6  o.Order_Date
  7  FROM Customer c
  8  RIGHT JOIN Orders o
  9  ON c.Customer_ID = o.Customer_ID;

CUSTOMER_ID          FIRST_NAME           LAST_NAME            ORDER_ID
-------------------- -------------------- -------------------- ----------
ORDER_DAT
---------
CUST-101             Ananya               Roy                  ORD-502
05-MAR-26

CUST-101             Ananya               Roy                  ORD-501
01-MAR-26

CUST-102             Rohan                Mehta                ORD-503
10-MAR-26


CUSTOMER_ID          FIRST_NAME           LAST_NAME            ORDER_ID
-------------------- -------------------- -------------------- ----------
ORDER_DAT
---------
CUST-103             Priya                Nair                 ORD-504
12-MAR-26

CUST-105             Aarav                Sharma               ORD-505
15-MAR-26

CUST-106             Ishita               Menon                ORD-506
16-MAR-26


CUSTOMER_ID          FIRST_NAME           LAST_NAME            ORDER_ID
-------------------- -------------------- -------------------- ----------
ORDER_DAT
---------
CUST-107             Kabir                Singh                ORD-507
17-MAR-26

CUST-108             Diya                 Patel                ORD-508
18-MAR-26

CUST-109             Arjun                Nair                 ORD-509
19-MAR-26
```

```
CUSTOMER_ID          FIRST_NAME           LAST_NAME            ORDER_ID
-------------------- -------------------- -------------------- ----------
ORDER_DAT
---------
CUST-110             Meera                Rao                  ORD-510
20-MAR-26

CUST-111             Rohan                Verma                ORD-511
21-MAR-26

CUST-112             Anika                Iyer                 ORD-512
22-MAR-26


CUSTOMER_ID          FIRST_NAME           LAST_NAME            ORDER_ID
-------------------- -------------------- -------------------- ----------
ORDER_DAT
---------
CUST-113             Nikhil               Bose                 ORD-513
23-MAR-26

CUST-114             Pooja                Reddy                ORD-514
24-MAR-26

CUST-115             Vikram               Kapoor               ORD-515
25-MAR-26


CUSTOMER_ID          FIRST_NAME           LAST_NAME            ORDER_ID
-------------------- -------------------- -------------------- ----------
ORDER_DAT
---------
CUST-116             Sneha                Gupta                ORD-516
26-MAR-26

CUST-117             Aditya               Mishra               ORD-517
27-MAR-26

CUST-118             Kavya                Das                  ORD-518
28-MAR-26


CUSTOMER_ID          FIRST_NAME           LAST_NAME            ORDER_ID
-------------------- -------------------- -------------------- ----------
ORDER_DAT
---------
CUST-119             Rahul                Joseph               ORD-519
29-MAR-26

CUST-120             Neha                 Varma                ORD-520
30-MAR-26


20 rows selected.
```

3.2 Find orders where customer details are missing

```
SQL> SELECT
  2  o.Order_ID,
  3  o.Order_Date
  4  FROM Customer c
  5  RIGHT JOIN Orders o
  6  ON c.Customer_ID = o.Customer_ID
  7  WHERE c.Customer_ID IS NULL;

no rows selected
```

3.3 Display all payment records with order details

```
SQL> SELECT
  2  p.Payment_ID,
  3  p.Payment_Mode,
  4  p.Payment_Status,
  5  o.Order_ID,
  6  o.Order_Date,
  7  o.Total_Amount
  8  FROM Orders o
  9  RIGHT JOIN Payment p
 10  ON o.Order_ID = p.Order_ID;

PAYMENT PAYMENT_MODE    PAYMENT_ST ORDER_ID   ORDER_DAT TOTAL_AMOUNT
------- --------------- ---------- ---------- --------- ------------
PAY001  Credit Card     Completed  ORD-501    01-MAR-26         1098
PAY002  UPI             Completed  ORD-502    05-MAR-26         1250
PAY003  Debit Card      Pending    ORD-503    10-MAR-26          599
PAY004  Net Banking     Completed  ORD-504    12-MAR-26         1149
PAY005  UPI             Completed  ORD-505    15-MAR-26          699
PAY006  Credit Card     Pending    ORD-506    16-MAR-26          799
PAY007  UPI             Completed  ORD-507    17-MAR-26          499
PAY008  Debit Card      Completed  ORD-508    18-MAR-26          599
PAY009  Net Banking     Completed  ORD-509    19-MAR-26          450
PAY010  UPI             Pending    ORD-510    20-MAR-26          650
PAY011  Credit Card     Completed  ORD-511    21-MAR-26          999

PAYMENT PAYMENT_MODE    PAYMENT_ST ORDER_ID   ORDER_DAT TOTAL_AMOUNT
------- --------------- ---------- ---------- --------- ------------
PAY012  Debit Card      Completed  ORD-512    22-MAR-26          850
PAY013  UPI             Pending    ORD-513    23-MAR-26          550
PAY014  Credit Card     Completed  ORD-514    24-MAR-26          799
PAY015  Net Banking     Completed  ORD-515    25-MAR-26          899
PAY016  UPI             Failed     ORD-516    26-MAR-26          450
PAY017  Credit Card     Completed  ORD-517    27-MAR-26          699
PAY018  Debit Card      Completed  ORD-518    28-MAR-26          750
PAY019  UPI             Pending    ORD-519    29-MAR-26          399
PAY020  Net Banking     Completed  ORD-520    30-MAR-26          550

20 rows selected.
```

4. RETRIEVE COMPLETE ORDER DETAILS

4.1 Generate a complete order report by combining:

- Customer Name
- Product Name
- Quantity
- Order Date
- Total Amount
- Payment Status

```
SQL> SELECT
  2  c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  3  b.Title AS Product_Name,
  4  od.Quantity,
  5  o.Order_Date,
  6  o.Total_Amount,
  7  p.Payment_Status
  8  FROM Customer c
  9  INNER JOIN Orders o
 10  ON c.Customer_ID = o.Customer_ID
 11  INNER JOIN Order_Details od
 12  ON o.Order_ID = od.Order_ID
 13  INNER JOIN Book b
 14  ON od.Product_ID = b.Book_ID
 15  INNER JOIN Payment p
 16  ON o.Order_ID = p.Order_ID;

CUSTOMER_NAME
----------------------------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_ST
---------- --------- ------------ ----------
Priya Nair
The Great Gatsby
         1 12-MAR-26         1149 Completed

Ananya Roy
The Great Gatsby
         1 01-MAR-26         1098 Completed

CUSTOMER_NAME
----------------------------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_ST
---------- --------- ------------ ----------

Rohan Mehta
Dune
         1 10-MAR-26          599 Pending

Ananya Roy
Dune

CUSTOMER_NAME
----------------------------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_ST
---------- --------- ------------ ----------
         1 01-MAR-26         1098 Completed

Priya Nair
Foundation
```

```
Foundation
         1 12-MAR-26         1149 Completed

Ananya Roy

CUSTOMER_NAME
----------------------------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_ST
---------- --------- ------------ ----------
Designing Data-Intensive Applications
         1 05-MAR-26         1250 Completed

Aarav Sharma
The Silent Patient
         1 15-MAR-26          699 Completed


CUSTOMER_NAME
----------------------------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_ST
---------- --------- ------------ ----------
Ishita Menon
Gone Girl
         1 16-MAR-26          799 Pending

Kabir Singh
Sherlock Holmes
         1 17-MAR-26          499 Completed

CUSTOMER_NAME
----------------------------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_ST
---------- --------- ------------ ----------

Diya Patel
Murder on the Orient Express
         1 18-MAR-26          599 Completed

Arjun Nair
Pride and Prejudice

CUSTOMER_NAME
----------------------------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_ST
---------- --------- ------------ ----------
         1 19-MAR-26          450 Completed
```

```
Meera Rao
Me Before You
         1 20-MAR-26          650 Pending

Rohan Verma

CUSTOMER_NAME
----------------------------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_ST
---------- --------- ------------ ----------
Harry Potter and the Philosopher's Stone
         1 21-MAR-26          999 Completed

Anika Iyer
The Hobbit
         1 22-MAR-26          850 Completed


CUSTOMER_NAME
----------------------------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_ST
---------- --------- ------------ ----------
Nikhil Bose
Dracula
         1 23-MAR-26          550 Pending

Pooja Reddy
The Shining
         1 24-MAR-26          799 Completed

CUSTOMER_NAME
----------------------------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_ST
---------- --------- ------------ ----------

Vikram Kapoor
Wings of Fire
         1 25-MAR-26          899 Completed

Sneha Gupta
The Diary of a Young Girl
```

```
CUSTOMER_NAME
----------------------------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_ST
---------- --------- ------------ ----------
         1 26-MAR-26          450 Failed

Rahul Joseph
Treasure Island
         1 29-MAR-26          399 Pending

Neha Varma

CUSTOMER_NAME
----------------------------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_ST
---------- --------- ------------ ----------
Around the World in 80 Days
         1 30-MAR-26          550 Completed

Aditya Mishra
Percy Jackson and the Lightning Thief
         1 27-MAR-26          699 Completed


CUSTOMER_NAME
----------------------------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_ST
---------- --------- ------------ ----------
Kavya Das
Angels and Demons
         1 28-MAR-26          750 Completed


22 rows selected.
```

5. DISPLAY CUSTOMER PURCHASE HISTORY

5.1 All products purchased by a customer

```
SQL> SELECT
  2  c.Customer_ID,
  3  c.First_Name,
  4  b.Title
  5  FROM Customer c
  6  INNER JOIN Orders o
  7  ON c.Customer_ID = o.Customer_ID
  8  INNER JOIN Order_Details od
  9  ON o.Order_ID = od.Order_ID
 10  INNER JOIN Book b
 11  ON od.Product_ID = b.Book_ID
 12  WHERE c.Customer_ID = 'CUST-101';

CUSTOMER_ID          FIRST_NAME
-------------------- --------------------
TITLE
--------------------------------------------------------------------------------
CUST-101             Ananya
The Great Gatsby

CUST-101             Ananya
Dune

CUST-101             Ananya
Designing Data-Intensive Applications
```

5.2 Total amount spent by each customer

```
SQL> SELECT
  2  c.Customer_ID,
  3  c.First_Name,
  4  SUM(o.Total_Amount) AS Total_Spent
  5  FROM Customer c
  6  INNER JOIN Orders o
  7  ON c.Customer_ID = o.Customer_ID
  8  GROUP BY c.Customer_ID,
  9  c.First_Name;

CUSTOMER_ID          FIRST_NAME           TOTAL_SPENT
-------------------- -------------------- -----------
CUST-101             Ananya                      2348
CUST-105             Aarav                        699
CUST-107             Kabir                        499
CUST-110             Meera                        650
CUST-113             Nikhil                       550
CUST-114             Pooja                        799
CUST-112             Anika                        850
CUST-116             Sneha                        450
CUST-119             Rahul                        399
CUST-102             Rohan                        599
CUST-103             Priya                       1149

CUSTOMER_ID          FIRST_NAME           TOTAL_SPENT
-------------------- -------------------- -----------
CUST-115             Vikram                       899
CUST-118             Kavya                        750
CUST-106             Ishita                       799
CUST-108             Diya                         599
CUST-120             Neha                         550
CUST-109             Arjun                        450
CUST-111             Rohan                        999
CUST-117             Aditya                       699

19 rows selected.
```

5.3 Number of orders placed by each customer

```
SQL> SELECT
  2  c.Customer_ID,
  3  c.First_Name,
  4  COUNT(o.Order_ID) AS Total_Orders
  5  FROM Customer c
  6  LEFT JOIN Orders o
  7  ON c.Customer_ID = o.Customer_ID
  8  GROUP BY c.Customer_ID,
  9  c.First_Name;

CUSTOMER_ID          FIRST_NAME           TOTAL_ORDERS
-------------------- -------------------- ------------
CUST-101             Ananya                          2
CUST-105             Aarav                           1
CUST-107             Kabir                           1
CUST-110             Meera                           1
CUST-113             Nikhil                          1
CUST-114             Pooja                           1
CUST-112             Anika                           1
CUST-116             Sneha                           1
CUST-119             Rahul                           1
CUST-102             Rohan                           1
CUST-103             Priya                           1

CUSTOMER_ID          FIRST_NAME           TOTAL_ORDERS
-------------------- -------------------- ------------
CUST-115             Vikram                          1
CUST-118             Kavya                           1
CUST-104             Vikram                          0
CUST-106             Ishita                          1
CUST-108             Diya                            1
CUST-120             Neha                            1
CUST-109             Arjun                           1
CUST-111             Rohan                           1
CUST-117             Aditya                          1

20 rows selected.
```

5.4 Latest purchase details of customers

```
SQL> SELECT
  2  c.Customer_ID,
  3  c.First_Name,
  4  MAX(o.Order_Date) AS Latest_Purchase
  5  FROM Customer c
  6  INNER JOIN Orders o
  7  ON c.Customer_ID = o.Customer_ID
  8  GROUP BY c.Customer_ID,
  9  c.First_Name;

CUSTOMER_ID          FIRST_NAME           LATEST_PU
-------------------- -------------------- ---------
CUST-101             Ananya               05-MAR-26
CUST-105             Aarav                15-MAR-26
CUST-107             Kabir                17-MAR-26
CUST-110             Meera                20-MAR-26
CUST-113             Nikhil               23-MAR-26
CUST-114             Pooja                24-MAR-26
CUST-112             Anika                22-MAR-26
CUST-116             Sneha                26-MAR-26
CUST-119             Rahul                29-MAR-26
CUST-102             Rohan                10-MAR-26
CUST-103             Priya                12-MAR-26

CUSTOMER_ID          FIRST_NAME           LATEST_PU
-------------------- -------------------- ---------
CUST-115             Vikram               25-MAR-26
CUST-118             Kavya                28-MAR-26
CUST-106             Ishita               16-MAR-26
CUST-108             Diya                 18-MAR-26
CUST-120             Neha                 30-MAR-26
CUST-109             Arjun                19-MAR-26
CUST-111             Rohan                21-MAR-26
CUST-117             Aditya               27-MAR-26

19 rows selected.
```

6. GENERATE MULTI-TABLE BUSINESS REPORTS

6.1 Report 1: Customer Order Report

- **Display Customer Name**
- **Order ID**
- **Order Date**
- **Order Status**

```
SQL> SELECT
  2  c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  3  o.Order_ID,
  4  o.Order_Date,
  5  o.Order_Status
  6  FROM Customer c
  7  INNER JOIN Orders o
  8  ON c.Customer_ID = o.Customer_ID;

CUSTOMER_NAME                          ORDER_ID   ORDER_DAT
-------------------------------------- ---------- ---------
ORDER_STATUS
--------------------
Ananya Roy                             ORD-501    01-MAR-26
Delivered

Ananya Roy                             ORD-502    05-MAR-26
Delivered

Rohan Mehta                            ORD-503    10-MAR-26
Pending


CUSTOMER_NAME                          ORDER_ID   ORDER_DAT
-------------------------------------- ---------- ---------
ORDER_STATUS
--------------------
Priya Nair                             ORD-504    12-MAR-26
Shipped

Aarav Sharma                           ORD-505    15-MAR-26
Delivered

Ishita Menon                           ORD-506    16-MAR-26
Pending


CUSTOMER_NAME                          ORDER_ID   ORDER_DAT
-------------------------------------- ---------- ---------
ORDER_STATUS
--------------------
Kabir Singh                            ORD-507    17-MAR-26
Shipped

Diya Patel                             ORD-508    18-MAR-26
Delivered

Arjun Nair                             ORD-509    19-MAR-26
Delivered
```

```
CUSTOMER_NAME                          ORDER_ID   ORDER_DAT
-------------------------------------- ---------- ---------
ORDER_STATUS
--------------------
Meera Rao                              ORD-510    20-MAR-26
Pending

Rohan Verma                            ORD-511    21-MAR-26
Delivered

Anika Iyer                             ORD-512    22-MAR-26
Shipped


CUSTOMER_NAME                          ORDER_ID   ORDER_DAT
-------------------------------------- ---------- ---------
ORDER_STATUS
--------------------
Nikhil Bose                            ORD-513    23-MAR-26
Pending

Pooja Reddy                            ORD-514    24-MAR-26
Delivered

Vikram Kapoor                          ORD-515    25-MAR-26
Delivered


CUSTOMER_NAME                          ORDER_ID   ORDER_DAT
-------------------------------------- ---------- ---------
ORDER_STATUS
--------------------
Sneha Gupta                            ORD-516    26-MAR-26
Cancelled

Aditya Mishra                          ORD-517    27-MAR-26
Delivered

Kavya Das                              ORD-518    28-MAR-26
Shipped


CUSTOMER_NAME                          ORDER_ID   ORDER_DAT
-------------------------------------- ---------- ---------
ORDER_STATUS
--------------------
Rahul Joseph                           ORD-519    29-MAR-26
Pending

Neha Varma                             ORD-520    30-MAR-26
Delivered


20 rows selected.
```

6.2 Report 2: Sales Report

- **Display Product Name**
- **Quantity Sold**
- **Total Revenue**

```
SQL> SELECT
  2  b.Title AS Product_Name,
  3  SUM(od.Quantity) AS Quantity_Sold,
  4  SUM(od.Quantity * od.Price) AS Total_Revenue
  5  FROM Book b
  6  INNER JOIN Order_Details od
  7  ON b.Book_ID = od.Product_ID
  8  GROUP BY b.Title;

PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------
Gone Girl
            1           799

Harry Potter and the Philosopher's Stone
            1           999

Dracula
            1           550


PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------
Dune
            2          1198

The Diary of a Young Girl
            1           450

Around the World in 80 Days
            1           550


PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------
The Silent Patient
            1           699

Sherlock Holmes
            1           499

Pride and Prejudice
            1           450
```

```
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------
Murder on the Orient Express
            1           599

Treasure Island
            1           399

Foundation
            1           650


PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------
Designing Data-Intensive Applications
            1          1250

The Great Gatsby
            2           998

Me Before You
            1           650


PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------
The Hobbit
            1           850

The Shining
            1           799

Wings of Fire
            1           899


PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------
Percy Jackson and the Lightning Thief
            1           699

Angels and Demons
            1           750


20 rows selected.
```

6.3 Report 3: Payment Analysis Report

- **Display Payment Mode**
- **Number of Transactions**
- **Successful Payments**

```
SQL> SELECT
  2  Payment_Mode,
  3  COUNT(*) AS Number_Of_Transactions,
  4  SUM(CASE
  5  WHEN Payment_Status = 'Completed'
  6  THEN 1
  7  ELSE 0
  8  END) AS Successful_Payments
  9  FROM Payment
 10  GROUP BY Payment_Mode;

PAYMENT_MODE    NUMBER_OF_TRANSACTIONS SUCCESSFUL_PAYMENTS
--------------- ---------------------- ------------------
Credit Card                          5                  4
Net Banking                          4                  4
UPI                                  7                  3
Debit Card                           4                  3
```

6.4 Report 4: Customer Purchase Analysis

6.4.1 Top Purchasing Customers

```
SQL> SELECT
  2  c.Customer_ID,
  3  c.First_Name,
  4  SUM(o.Total_Amount) AS Total_Spent
  5  FROM Customer c
  6  INNER JOIN Orders o
  7  ON c.Customer_ID = o.Customer_ID
  8  GROUP BY c.Customer_ID,
  9  c.First_Name
 10  ORDER BY Total_Spent DESC;

CUSTOMER_ID          FIRST_NAME           TOTAL_SPENT
-------------------- -------------------- -----------
CUST-101             Ananya                      2348
CUST-103             Priya                       1149
CUST-111             Rohan                        999
CUST-115             Vikram                       899
CUST-112             Anika                        850
CUST-106             Ishita                       799
CUST-114             Pooja                        799
CUST-118             Kavya                        750
CUST-117             Aditya                       699
CUST-105             Aarav                        699
CUST-110             Meera                        650

CUSTOMER_ID          FIRST_NAME           TOTAL_SPENT
-------------------- -------------------- -----------
CUST-108             Diya                         599
CUST-102             Rohan                        599
CUST-120             Neha                         550
CUST-113             Nikhil                       550
CUST-107             Kabir                        499
CUST-109             Arjun                        450
CUST-116             Sneha                        450
CUST-119             Rahul                        399

19 rows selected.
```

6.4.2 Customers with Maximum Orders

```
SQL> SELECT
  2  c.Customer_ID,
  3  c.First_Name,
  4  COUNT(o.Order_ID) AS Total_Orders
  5  FROM Customer c
  6  INNER JOIN Orders o
  7  ON c.Customer_ID = o.Customer_ID
  8  GROUP BY c.Customer_ID,
  9  c.First_Name
 10  ORDER BY Total_Orders DESC;

CUSTOMER_ID          FIRST_NAME           TOTAL_ORDERS
-------------------- -------------------- ------------
CUST-101             Ananya                          2
CUST-107             Kabir                           1
CUST-110             Meera                           1
CUST-113             Nikhil                          1
CUST-114             Pooja                           1
CUST-112             Anika                           1
CUST-116             Sneha                           1
CUST-119             Rahul                           1
CUST-102             Rohan                           1
CUST-103             Priya                           1
CUST-115             Vikram                          1

CUSTOMER_ID          FIRST_NAME           TOTAL_ORDERS
-------------------- -------------------- ------------
CUST-118             Kavya                           1
CUST-106             Ishita                          1
CUST-108             Diya                            1
CUST-120             Neha                            1
CUST-109             Arjun                           1
CUST-111             Rohan                           1
CUST-117             Aditya                          1
CUST-105             Aarav                           1

19 rows selected.

SQL>
```

6.4.3 Customers with Highest Spending

```
SQL> SELECT
  2  c.Customer_ID,
  3  c.First_Name,
  4  SUM(o.Total_Amount) AS Highest_Spending
  5  FROM Customer c
  6  INNER JOIN Orders o
  7  ON c.Customer_ID = o.Customer_ID
  8  GROUP BY c.Customer_ID,
  9  c.First_Name
 10  ORDER BY Highest_Spending DESC;

CUSTOMER_ID          FIRST_NAME           HIGHEST_SPENDING
-------------------- -------------------- ----------------
CUST-101             Ananya                           2348
CUST-103             Priya                            1149
CUST-111             Rohan                             999
CUST-115             Vikram                            899
CUST-112             Anika                             850
CUST-106             Ishita                            799
CUST-114             Pooja                             799
CUST-118             Kavya                             750
CUST-117             Aditya                            699
CUST-105             Aarav                             699
CUST-110             Meera                             650

CUSTOMER_ID          FIRST_NAME           HIGHEST_SPENDING
-------------------- -------------------- ----------------
CUST-108             Diya                              599
CUST-102             Rohan                             599
CUST-120             Neha                              550
CUST-113             Nikhil                            550
CUST-107             Kabir                             499
CUST-109             Arjun                             450
CUST-116             Sneha                             450
CUST-119             Rahul                             399

19 rows selected.
```